NEWTON SCHOOL

ALGEBRA 1 | ANSWER KEY

Radicals Two Ways — Answer Key

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Part A

1) $3\sqrt{3} - 2\sqrt{12} + 4\sqrt{\frac{1}{3}}$

$3\sqrt{3} - 2(2\sqrt{3}) + 4 \cdot \frac{1}{\sqrt{3}} = 3\sqrt{3} - 4\sqrt{3} + \frac{4\sqrt{3}}{3}$

$= (3 - 4 + \frac{4}{3})\sqrt{3} = \frac{1}{3}\sqrt{3}$

$\boxed{\frac{\sqrt{3}}{3}}$

2) $8\sqrt{10} - 3\sqrt{40} + 5\sqrt{\frac{1}{10}}$

$8\sqrt{10} - 6\sqrt{10} + \frac{5}{\sqrt{10}} = 2\sqrt{10} + \frac{\sqrt{10}}{2}$

$\boxed{\frac{5\sqrt{10}}{2}}$

3) $2\sqrt{\frac{7}{2}} + 4\sqrt{\frac{7}{8}} - \frac{1}{2}\sqrt{98}$

$2 \cdot \frac{\sqrt{14}}{2} + 4 \cdot \frac{\sqrt{14}}{4} - \frac{7\sqrt{2}}{2} = 2\sqrt{14} - \frac{7\sqrt{2}}{2}$

$\boxed{\frac{4\sqrt{14} - 7\sqrt{2}}{2}}$

4) $3\sqrt{\frac{5}{12}} + \sqrt{\frac{12}{5}} - \frac{1}{3}\sqrt{60}$ two-way!

$\frac{\sqrt{15}}{2} + \frac{2\sqrt{15}}{5} - \frac{2\sqrt{15}}{3}$ (LCD 30)

$= \frac{15\sqrt{15}+12\sqrt{15}-20\sqrt{15}}{30}$

$\boxed{\frac{7\sqrt{15}}{30}}$

5) $5\sqrt{3}(\sqrt{6} + 2\sqrt{8})$

$5\sqrt{18} + 10\sqrt{24} = 5(3\sqrt{2}) + 10(2\sqrt{6})$

$\boxed{15\sqrt{2} + 20\sqrt{6}}$

6) $5\sqrt{2}(4\sqrt{8} - 2\sqrt{12})$

$20\sqrt{16} - 10\sqrt{24} = 20(4) - 10(2\sqrt{6})$

$\boxed{80 - 20\sqrt{6}}$

7) $2\sqrt{49x^3} - 3\sqrt{16x^3}$

$2 \cdot 7x\sqrt{x} - 3 \cdot 4x\sqrt{x} = 14x\sqrt{x} - 12x\sqrt{x}$

$\boxed{2x\sqrt{x}}$

8) $4\sqrt{72s^4} - 2s\sqrt{200s^2}$

$4 \cdot s^2 \cdot 6\sqrt{2} - 2s \cdot s \cdot 10\sqrt{2} = 24s^2\sqrt{2} - 20s^2\sqrt{2}$

$\boxed{4s^2\sqrt{2}}$

9) $\sqrt{\frac{x^2}{16} + \frac{x^2}{25}}$

$\sqrt{\frac{25x^2+16x^2}{400}} = \sqrt{\frac{41x^2}{400}} = \frac{x\sqrt{41}}{20}$

$\boxed{\frac{x\sqrt{41}}{20}}$

10) $\sqrt{\frac{x^2}{49} - \frac{x^2}{121}}$

$\sqrt{\frac{121x^2-49x^2}{5929}} = \sqrt{\frac{72x^2}{5929}} = \frac{x \cdot 6\sqrt{2}}{77}$

$\boxed{\frac{6x\sqrt{2}}{77}}$

Part B

11) $\sqrt{\frac{x^2}{a^2}+\frac{x^2}{b^2}}$

$\sqrt{\frac{b^2x^2+a^2x^2}{a^2b^2}} = \frac{x}{ab}\sqrt{a^2+b^2}$

$\boxed{\frac{x\sqrt{a^2+b^2}}{ab}}$

12) $\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}$ two-way!

Way 1: $\frac{\sqrt{2}}{2}+\frac{\sqrt{3}}{3} = \frac{3\sqrt{2}+2\sqrt{3}}{6}$

$\boxed{\frac{3\sqrt{2}+2\sqrt{3}}{6}}$

13) $\frac{3}{\sqrt{5}}-\frac{2}{\sqrt{15}}$

$\frac{3\sqrt{5}}{5}-\frac{2\sqrt{15}}{15} = \frac{9\sqrt{5}-2\sqrt{15}}{15}$

$\boxed{\frac{9\sqrt{5}-2\sqrt{15}}{15}}$

14) $\sqrt{\frac{2}{3}}+\sqrt{\frac{3}{2}}$ two-way!

$\frac{\sqrt{6}}{3}+\frac{\sqrt{6}}{2} = \frac{2\sqrt{6}+3\sqrt{6}}{6}$

$\boxed{\frac{5\sqrt{6}}{6}}$

15) $2\sqrt{\frac{1}{8}}+\sqrt{\frac{1}{2}}$

$\frac{2}{2\sqrt{2}}+\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}}$

$\boxed{\sqrt{2}}$

16) $\sqrt{8}+\sqrt{18}-\sqrt{2}$

$2\sqrt{2}+3\sqrt{2}-\sqrt{2}$

$\boxed{4\sqrt{2}}$

17) $\sqrt{20}-\sqrt{45}+\sqrt{80}$

$2\sqrt{5}-3\sqrt{5}+4\sqrt{5}$

$\boxed{3\sqrt{5}}$

18) $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})$

Conjugates: $3-2$

$\boxed{1}$

19) $(2\sqrt{5}+\sqrt{3})^2$

$(2\sqrt{5})^2+2(2\sqrt{5})(\sqrt{3})+(\sqrt{3})^2 = 20+4\sqrt{15}+3$

$\boxed{23+4\sqrt{15}}$

★ 20) $4\sqrt{\frac{5}{6}}-\sqrt{\frac{3}{10}}$ (the Kenny problem)

WAY 1 — Rationalize each FIRST, then combine

$\frac{4\sqrt{5}}{\sqrt{6}}\cdot\frac{\sqrt{6}}{\sqrt{6}}-\frac{\sqrt{3}}{\sqrt{10}}\cdot\frac{\sqrt{10}}{\sqrt{10}}$

$= \frac{4\sqrt{30}}{6}-\frac{\sqrt{30}}{10} = \frac{2\sqrt{30}}{3}-\frac{\sqrt{30}}{10}$

LCD 30: $\frac{20\sqrt{30}-3\sqrt{30}}{30} = \frac{17\sqrt{30}}{30}$

WAY 2 — Build $\sqrt{30}$ common denom FIRST, rationalize LAST

$\frac{4\sqrt{5}}{\sqrt{6}}\cdot\frac{\sqrt{5}}{\sqrt{5}}-\frac{\sqrt{3}}{\sqrt{10}}\cdot\frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{20}{\sqrt{30}}-\frac{3}{\sqrt{30}} = \frac{17}{\sqrt{30}}$

$= \frac{17}{\sqrt{30}}\cdot\frac{\sqrt{30}}{\sqrt{30}} = \frac{17\sqrt{30}}{30}$

$\boxed{\frac{17\sqrt{30}}{30}}$ *Both ways → same answer ✓*